SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	03-08-2026	Duration:	60 minutes

Code of Conduct

*I **K Madan Mohan Reddy (192411206)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___**Byreddy**______________________

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

"Online Loan Eligibility Prediction and Approval Management Platform"

Study and Analyze the Scenario

- **Carefully understand the given problem statement.**

  The Online Loan Eligibility Prediction and Approval Management Platform is a web-based system designed to simplify and automate the loan application and approval process for both customers and financial institutions. The platform enables users to register securely, submit loan applications, upload required documents, and track the status of their applications online. It analyzes applicant information such as age, income, employment status, credit history, loan amount, and repayment capacity to predict loan eligibility using predefined business rules or machine learning techniques.

  - Loan officers and administrators can review applications, verify documents, approve or reject loans, and generate reports through a centralized dashboard.
  - By reducing manual intervention, minimizing processing time, improving decision accuracy, and ensuring transparency throughout the loan approval process, the platform enhances customer satisfaction while increasing the operational efficiency of financial institutions.

- **Identify the business domain, stakeholders, existing challenges, and system objectives.**
- **Business Domain**

The Online Loan Eligibility Prediction and Approval Management Platform belongs to the Banking, Financial Services, and Insurance (BFSI) domain. It focuses on digital loan processing by automating loan eligibility prediction, application management, document verification, and approval workflows to improve efficiency and customer experience.

- **Stakeholders**

  - Customers/Applicants: Register, apply for loans, upload documents, and track application status.
  - Loan Officers: Verify customer information, review applications, and recommend approvals or rejections.
  - Administrators: Manage users, monitor loan applications, maintain system records, and generate reports.
  - Bank/Financial Institution: Provides loan services, defines eligibility criteria, and oversees the approval process.
  - Credit Verification Agencies: Supply credit history and credit score information to support eligibility assessment**.**

- **Existing Challenges**

  - Manual loan processing leads to delays in approval.
  - Large volumes of applications increase workload for staff.
  - Human errors may occur during eligibility assessment.
  - Lack of transparency makes it difficult for applicants to track application status.
  - Paper-based documentation is time-consuming and difficult to manage.
  - Inconsistent decision-making may affect fairness and accuracy.
  - System Objectives
  - Automate loan eligibility prediction using customer financial data.
  - Enable secure online registration and loan application submission.
  - Simplify document upload and verification.
  - Provide faster and more accurate loan approval decisions.
  - Allow applicants to track their loan status in real time.
  - Improve operational efficiency while reducing manual effort and processing time.
  - Generate reports and maintain a centralized database for better management and auditing.

2. **Identify Functional and Non-Functional Requirements**
   - **List all functional requirements required by the proposed system.**

The Online Loan Eligibility Prediction and Approval Management Platform shall provide all the essential functionalities required to automate the loan application and approval process. The system shall allow customers to register, log in securely, manage their profiles, submit loan applications, and upload the necessary supporting documents. It shall analyze applicant information such as income, employment status, credit history, age, and loan amount to predict loan eligibility using predefined rules or machine learning techniques. Loan officers shall verify the submitted documents and review applications before approving or rejecting them.

- Customers shall be able to track the status of their applications and receive notifications regarding approvals, rejections, or requests for additional information.
- The platform shall also provide administrators with features to manage users, monitor loan applications, generate reports, maintain loan records, enforce role-based access control, and securely store all application data, ensuring an efficient, transparent, and reliable loan management process.

- **Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.**

The Online Loan Eligibility Prediction and Approval Management Platform shall satisfy several non-functional requirements to ensure high-quality system performance and user satisfaction. The system shall provide high performance by processing loan applications and predicting eligibility within a few seconds under normal operating conditions. It shall ensure security through secure user authentication, role-based access control, data encryption, and protection against unauthorized access. The platform shall maintain reliability by providing accurate loan eligibility predictions, minimizing system failures, and preserving data integrity. It shall offer usability through a simple, intuitive, and responsive user interface that enables customers and administrators to perform tasks with minimal effort.

- The system shall be scalable to accommodate an increasing number of users, loan applications, and data without degrading performance.
- It shall support maintainability by using a modular architecture, well-documented code, and easy software updates for future enhancements.
- Additionally, the platform shall ensure high availability by remaining accessible with minimal downtime, supporting continuous operation, regular backups, and efficient recovery mechanisms to guarantee uninterrupted loan services.

3. **Identify Actors and Use Cases**
   - **Identify all primary and secondary actors interacting with the system.**

The Online Loan Eligibility Prediction and Approval Management Platform involves several actors who interact with the system to perform different tasks. The primary actors include Customers (Applicants), who register, log in, submit loan applications, upload required documents, and track their application status; Loan Officers, who verify applicant details, review documents, evaluate loan applications, and approve or reject loans; and the Administrator, who manages users, configures system settings, monitors loan processing, generates reports, and maintains the overall platform. The secondary actors include Credit Verification Agencies, which provide credit scores and credit history for eligibility assessment; Notification Services (Email/SMS), which send application updates and approval or rejection notifications to users; Payment Gateway/Banking Systems, which facilitate loan disbursement and payment-related transactions; and the Database Server, which securely stores and retrieves customer information, loan records, and system data.

- **List the major use cases associated with each actor.**

  The major use cases of the Online Loan Eligibility Prediction and Approval Management Platform are associated with different actors based on their responsibilities. Customers (Applicants) can register, log in, manage their profiles, apply for loans, upload required documents, check loan eligibility, track application status, receive notifications, and log out. Loan Officers can log in, review loan applications, verify customer documents, evaluate eligibility results, approve or reject loan requests, update application status, and generate loan-related reports. The Administrator can log in, manage user accounts, assign roles, monitor all loan applications, manage loan policies, generate system reports, maintain the database, and oversee the overall system operations. Credit Verification Agencies provide applicants' credit scores and credit history for eligibility verification.

  - Notification Services send email or SMS alerts regarding application submission, approval, rejection, and status updates.
  - Payment Gateway/Banking Systems process loan disbursement and repayment transactions, while the Database Server stores and retrieves customer details, loan applications, documents, and transaction records securely.

- Draw a Use Case Diagram representing the interactions between actors and the system.

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+----------------------------------------------+
| Online Loan Eligibility Prediction and       |
| Approval Management Platform                 |
+----------------------------------------------+


Customer (Applicant)
     |
```

```
        |--------------------> (Register)

        |--------------------> (Login)

        |--------------------> (Manage Profile)

        |--------------------> (Apply for Loan)

        |--------------------> (Upload Documents)

        |--------------------> (Check Loan Eligibility)

        |--------------------> (Track Application Status)

        |--------------------> (Receive Notifications)

        |--------------------> (Logout)



Loan Officer

        |

        |--------------------> (Login)

        |--------------------> (Verify Documents)

        |--------------------> (Review Loan Application)

        |--------------------> (Approve/Reject Loan)

        |--------------------> (Update Application Status)

        |--------------------> (Generate Reports)



Administrator

        |

        |--------------------> (Login)

        |--------------------> (Manage Users)
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|--------------------> (Manage Loan Policies)

|--------------------> (Monitor Applications)

|--------------------> (Generate Reports)

|--------------------> (Maintain Database)

|--------------------> (Logout)



Credit Verification Agency

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|--------------------> (Provide Credit Score)



Notification Service

|

|--------------------> (Send Email/SMS Alerts)



Banking/Payment System

|

|--------------------> (Disburse Approved Loan)
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4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction

     The Online Loan Eligibility Prediction and Approval Management Platform is a web-based application that automates the loan application and approval process. It enables customers to apply for loans online, upload required documents, and receive quick eligibility predictions based on their financial information. The platform also assists loan officers and administrators in verifying applications, making approval decisions, and managing loan records efficiently. The system aims to reduce manual effort, improve decision accuracy, and provide a transparent and secure loan processing experience.

- Problem Description

  Traditional loan approval systems are often manual, time-consuming, and prone to human errors. Applicants must visit bank branches, submit paper documents, and wait several days or weeks for approval. Financial institutions also face difficulties in processing a large number of applications efficiently. The proposed system addresses these challenges by automating loan eligibility prediction, document verification, application tracking, and approval management through a centralized online platform.

- Scope of the System

  The system allows customers to register, log in, apply for loans, upload documents, check eligibility, and track application status. Loan officers can verify documents, review applications, and approve or reject loans. Administrators can manage users, monitor loan processing, generate reports, and maintain system data. The platform supports secure, reliable, and efficient loan management for banks and financial institutions.

Functional Requirements

- User registration and secure login.
- Profile management.
- Online loan application submission.
- Document upload and verification.
- Loan eligibility prediction.
- Loan application review.
- Loan approval or rejection.
- Application status tracking.
- Email/SMS notifications.
- User and role management.
- Report generation.
- Search and filter functionality.
- Loan record management.
- Audit log maintenance.

- Secure logout.

Non-Functional Requirements

- Performance: Process loan applications and eligibility predictions quickly.
- Security: Secure authentication, authorization, encryption, and role-based access control.
- Reliability: Accurate processing with minimal system failures.
- Usability: User-friendly and responsive interface.
- Scalability: Support increasing users and loan applications.
- Maintainability: Modular design for easy updates and maintenance.
- Availability: High system availability with backup and recovery mechanisms

  - Data Requirements

  The system stores customer details, loan applications, uploaded documents, eligibility results, loan status, and user records securely in the database.

  - External Interface Requirements

  The system provides a responsive web interface, supports major web browsers, uses a MySQL database, integrates with email/SMS services, and communicates securely through HTTPS.

5. System Features

- User Registration and Login
- Online Loan Application
- Loan Eligibility Prediction
- Document Upload
- Loan Approval/Rejection
- Application Status Tracking
- Notifications
- Report Generation
- User Management

6. **Identify Assumptions and Constraints**

List the assumptions considered while developing the system

- Users have a stable internet connection to access the platform.
- Customers provide accurate and complete personal and financial information.
- All required documents are uploaded in the prescribed format.
- Loan officers and administrators have valid system credentials.
- The eligibility prediction model is trained with reliable and up-to-date data.
- The system database is available and securely stores all user and loan information.
- External services such as email/SMS notifications and credit verification are available when required.

Identify technical, operational, economic, legal, and time-related constraints affecting the project.

- Technical Constraints: The system requires a stable internet connection, compatible web browsers, secure database management, and reliable server infrastructure.
- Operational Constraints: Users must provide accurate information, and loan officers must verify applications according to the bank's policies.
- Economic Constraints: The project must be completed within the allocated budget, including software development, deployment, and maintenance costs.
- Legal Constraints: The system must comply with banking regulations, data privacy laws, and information security standards to protect customer data.
- Time-Related Constraints: The project must be completed within the planned schedule, with all modules developed, tested, and deployed before the project deadline.

7. **Validate Requirement Completeness and Consistency**
   - **Verify whether all stakeholder requirements have been addressed.**

All identified stakeholder requirements have been addressed in the proposed system. Customers can securely register, apply for loans, upload documents, track application status, and receive notifications. Loan officers can verify applications, review documents, and approve or reject loan requests. Administrators can manage users, monitor loan processing, generate reports, and maintain the system. The platform also supports secure data management, role-based access control, and loan eligibility prediction, ensuring that the functional and non-functional requirements of all stakeholders are satisfied.

- **Check for ambiguity, redundancy, inconsistency, and missing requirements**.

The requirements have been reviewed for quality and completeness. Ambiguity has been minimized by using clear and specific requirement statements. Redundancy has been eliminated by ensuring that each requirement is defined only once. Consistency has been maintained so that all functional and non-functional requirements align with the system objectives and stakeholder needs. Missing requirements have been addressed by including user authentication, loan application, eligibility prediction, document verification, loan approval, status tracking, notifications, reporting, security, and data management. Therefore, the proposed SRS is complete, consistent, and suitable for system development.

- **Suggest improvements to enhance the quality and completeness of the requirement specification.**

The quality and completeness of the requirement specification can be improved by adding detailed business rules for loan eligibility, clearly defining user roles and access permissions, specifying system performance and security requirements, including exception and error-handling scenarios, documenting data validation rules, and describing integration with external services such as credit verification and notification systems. Additionally, incorporating acceptance criteria, backup and recovery requirements, and future scalability considerations will make the SRS more comprehensive, reliable, and easier to implement and maintain.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5

Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25